DESIGN AND DEVELOPMENT OF A CONTEXT-AWARE SENTIMENT ANALYSIS SYSTEM USING MULTI-LEVEL NLP

ASSESSMENT REPORT

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GROUP 6 – VADER + TRANSFORMER HYBRID MULTI-LEVEL NLP APPROACH

Academic Project Report

Same Problem Statement as Group 1; distinct core approach: VADER + Transformer Hybrid

Group 6 preserves the same customer-review problem and five-unit NLP coverage while using VADER lexical sentiment, a transformer sentiment classifier, and contextual model-fusion rules as the sentiment core.

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Group 6 proposes a VADER + Transformer Hybrid Multi-Level NLP Sentiment Analysis System.

The system combines:

Regular expressions and preprocessing

Tokenization and Porter stemming

Unigram, bigram, and trigram features

POS features for aspect and opinion identification

CFG supporting syntax analysis

Context-aware negation handling

Semantic representations / FOPC

Model-assisted WSD using contextual evidence

Context-aware reference resolution

Discourse marker detection and weighting

Aspect identification and aspect-level sentiment

VADER lexical sentiment scoring

Transformer sentiment prediction using a pretrained text-classification model

Contextual fusion of VADER, transformer, aspect, negation, and discourse evidence

Natural-language generation of the final explanation

Group 6 therefore uses VADER and a transformer model as the central sentiment engines while retaining the multi-level NLP pipeline required by the problem statement.

1. Problem Formulation

Online customer reviews frequently contain multiple opinions within the same review. A review may contain positive opinions about one product aspect and negative opinions about another aspect. Simple sentiment analysis systems that classify text by counting positive and negative words may therefore produce incorrect results.

The problem considered in this assignment is the development of a Python-based context-aware sentiment analysis system that classifies a customer review as Positive, Negative, or Neutral by considering linguistic information at multiple levels.

The system must analyze morphology, N-grams, POS information, syntactic structure, negation, semantic meaning, word sense, reference expressions, discourse relations, product aspects, and overall sentiment.

Customer review used for analysis:

“I was expecting the phone to be amazing. The display is bright and the camera takes excellent pictures, but the battery is disappointing. It does not last long, and I am unhappy with it. However, the service team replaced it quickly, which was impressive. Overall, I am satisfied with the phone.”

The proposed Group 6 solution uses VADER + Transformer Hybrid as the core sentiment method. Instead of making an abrupt binary decision from isolated keywords, the system converts linguistic evidence into contextual degrees of positive, negative, mixed, and neutral support. These degrees are modified by negation, aspect context, reference resolution, and discourse markers before producing aspect-level and overall sentiment classifications.

Input

A natural-language customer review containing opinions about one or more product aspects.

Output

- Morphological analysis
- N-gram representations
- POS information
- Syntactic/CFG analysis
- Negation detection
- Semantic representation
- Word Sense Disambiguation
- Reference resolution
- Discourse relations
- Aspect identification
- Contextual sentiment sentiment evidence values
- Aspect-level sentiment
- Contextual-weighted overall sentiment score
- Final sentiment classification
- Natural-language explanation

Problem

How can the sentiment of a customer review be correctly determined when positive and negative opinions, negation, contrast, references, and multiple product aspects occur in the same text, while uncertainty and strength of evidence are represented gradually rather than by rigid rules?

Why Keyword-Based Analysis Is Insufficient

- “does not last long” expresses a negative opinion through negation.
- “but” changes the discourse relationship between camera/display and battery.
- “However” introduces a contrasting positive opinion about service.
- “it” refers to different entities depending on context.
- “Overall” indicates a global/final opinion.
- Different aspects of the same product have different sentiment polarities.
- A fixed positive/negative threshold cannot represent mixed or uncertain evidence well.

Therefore, a context-aware contextual approach is used to model sentiment as a degree of evidence and to preserve aspect-specific context before overall aggregation.

2. Objectives

1. To develop a contextual-logic-based multi-level NLP system for context-aware sentiment analysis.
2. To preprocess and tokenize customer reviews.
3. To perform morphological analysis and stemming.
4. To generate unigram, bigram, and trigram representations as supporting features.
5. To perform POS-based linguistic analysis.
6. To construct CFG rules for syntactic support.
7. To explicitly detect and scope negation.
8. To represent selected statements using First-Order Predicate Calculus.
9. To perform contextual Word Sense Disambiguation using contextual contextual evidence.
10. To resolve references such as "it" and "which" using contextual confidence.
11. To identify discourse relations such as contrast and conclusion.
12. To identify product aspects such as display, camera, battery, service, and phone.
13. To construct contextual sentiment features from lexical, N-gram, morphological, POS, and contextual evidence.
14. To model negation as a contextual rule that reduces or reverses sentiment strength for the affected aspect.
15. To weight discourse markers using contextual model fusion rules.
16. To calculate contextual aspect-level sentiment sentiment evidences.
17. To aggregate contextual aspect evidence into an overall sentiment score.
18. To classify the review as Positive, Negative, or Neutral.
19. To generate a natural-language explanation of the result.
20. To demonstrate why contextual context-aware sentiment analysis is more informative than simple keyword counting.

3. Expected Outcomes

- Correct tokenization of the review.
- Successful stemming of selected words.
- Generation of unigrams, bigrams, and trigrams.
- Identification of relevant POS information.
- Construction of suitable CFG rules.
- Identification of negative expressions.
- Recognition of negation scope.
- Identification of positive and negative aspects.
- Resolution of contextual references.
- Identification of discourse relations.
- Contextual sentiment sentiment evidence values for aspect evidence.
- Aspect-level sentiment classification.
- Contextual-weighted overall sentiment classification.
- Generation of a human-readable explanation.

Expected aspect-level results for the given review:

Aspect	**Expected Sentiment**	**Main Evidence**
Display	Positive	bright
Camera	Positive	excellent pictures
Battery	Negative	disappointing; does not last long; unhappy
Service	Positive	replaced it quickly; impressive
Overall Phone	Positive	Overall, I am satisfied with the phone

Expected final classification: OVERALL SENTIMENT: POSITIVE

4. Requirements

4.1 Functional Requirements

- Accept a customer review as input.
- Perform text preprocessing.
- Tokenize the input.
- Perform morphological analysis.
- Apply stemming.
- Generate unigrams, bigrams, and trigrams.
- Identify word classes and POS information.
- Apply CFG-based syntactic analysis.
- Detect explicit negation.
- Determine the scope of negation.
- Generate semantic representations for selected statements.
- Perform Word Sense Disambiguation.
- Resolve contextual references.
- Identify discourse markers.
- Identify product aspects.
- Generate contextual linguistic features.
- Calculate contextual sentiment sentiment evidence values.
- Apply contextual negation and context rules.
- Apply contextual discourse weighting.
- Calculate overall contextual sentiment.
- Generate a natural-language explanation.

4.2 Software Requirements

- Python 3.x
- NLTK
- Regular Expression library
- Standard Python math functions for contextual sentiment evidence
- VS Code / Jupyter Notebook / Google Colab

4.3 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic processor capable of running Python

5. Constraints

- The final result must not depend only on keyword frequency.
- Negation must be handled explicitly.
- Contrast markers must influence aspect interpretation.
- Aspect-level and overall sentiment must be distinguished.

- Ambiguous references must be handled using contextual rules.
- Contextual rules and sentiment evidence functions are manually designed for the academic prototype.
- The prototype is primarily designed for English customer reviews.
- The contextual lexicon may not cover every possible expression.
- Sarcasm and irony are outside the core implementation.
- The system is an academic prototype rather than a production-scale commercial system.

6. Assumptions

- The input review is written in English.
- The review is grammatically understandable.
- Product aspects can be identified using predefined terms and contextual rules.
- A manually designed sentiment lexicon is sufficient for demonstration.
- Sentiment-bearing words have interpretable polarity and strength.
- Discourse markers such as “but”, “However”, and “Overall” provide useful context.
- Pronouns can be resolved using nearby entities and aspect context.
- Contextual sentiment evidence functions can approximate sentiment strength for the demonstration.
- The overall sentiment can be estimated by combining aspect-level contextual evidence and explicit overall sentiment.
- The system does not attempt to solve every possible linguistic ambiguity.

7. Application of All Five Units

7.1 Unit I – Morphology

Morphological processing identifies the base or stemmed form of words. Group 6 uses Porter stemming during preprocessing, and the resulting stem information becomes a contextual feature: high agreement between a stem and a sentiment lexicon entry increases the sentiment evidence degree of that sentiment feature.

Word	**Stem**	Contextual Feature Role
amazing	amaz	Transformer/VADER lexical feature
pictures	pictur	Context feature
disappointing	disappoint	Negative lexical evidence
unhappy	unhappi	Negative context feature
replaced	replac	Service-event feature
impressive	impress	Positive model evidence
satisfied	satisfi	Strong global positive evidence

Conceptual finite-state processing: Input Word → Lexical Form → Morphological Rule → Stem → Contextual Feature.

7.2 Unit II – N-Grams and POS

N-grams are supporting contextual features in Group 6. Unigrams provide individual evidence, bigrams capture local associations, and trigrams preserve phrases whose combined meaning is stronger than isolated words.

Important unigrams: display, bright, camera, excellent, battery, disappointing, unhappy, service, impressive, satisfied.

Important bigrams: display bright, excellent pictures, battery disappointing, does not, last long, service team, replaced quickly.

Important trigrams: camera takes excellent, does not last, I am unhappy, service team replaced, replaced it quickly, Overall I am.

POS information separates likely aspect nouns from opinion-bearing adjectives and verbs. POS confidence is represented as a contextual feature rather than as an absolute requirement.

Token	**Role**	Contextual Use
display	NOUN	Aspect feature
bright	ADJECTIVE	Positive opinion feature
camera	NOUN	Aspect feature
excellent	ADJECTIVE	Strong positive feature
battery	NOUN	Aspect feature
disappointing	ADJECTIVE	Strong negative feature
service	NOUN	Aspect/event feature
impressive	ADJECTIVE	Positive opinion sentiment evidence
satisfied	ADJECTIVE	Global positive sentiment evidence
it	PRONOUN	Reference-context feature

7.3 Unit III – Context-Free Grammar

CFG is used as supporting syntax. The grammar below provides structural evidence that can increase the contextual confidence that an adjective or predicate describes a nearby aspect.

```
S → NP VP
VP → V NP
VP → V ADJP
NP → DET N
NP → DET N PP
PP → PREP NP
ADJP → ADV ADJ
ADJP → ADJ
```

For “The display is bright”, a simplified parse is:

```
S
├── NP
│   ├── DET → The
│   └── N → display
└── VP
    ├── V → is
    └── ADJP
        └── ADJ → bright
```

CFG evidence is used as a supporting contextual sentiment evidence for aspect-opinion attachment; it is not itself the final sentiment classifier.

7.4 Negation Scope

The phrase “It does not last long” contains the negation marker not. Group 6 represents negation with a contextual rule: when negation sentiment evidence is high and a positive/neutral predicate falls inside the detected scope, its positive sentiment evidence is reduced and negative sentiment evidence is increased. After reference resolution, this contextual adjustment is applied specifically to the battery aspect.

7.5 Unit IV – Semantic Analysis

Selected statements are represented using simplified FOPC-style predicates. These semantic forms become interpretable evidence objects for the contextual system.

Statement	**Semantic Representation**
The display is bright	Bright(Display)
The camera takes excellent pictures	Excellent(CameraPictures)
The battery is disappointing	Disappointing(Battery)
The battery does not last long	Negative(BatteryPerformance)
The service replacement was impressive	Impressive(ServiceReplacement)
I am satisfied with the phone	Satisfied(User, Phone)

7.6 Word Sense Disambiguation

WSD is implemented using contextual contextual sentiment evidence. For example, “bright” can have several meanings. When it occurs near “display”, the sentiment evidence of the display-brightness sense becomes high. “Service” also receives high customer-support sense sentiment evidence because it occurs with “service team” and product replacement.

Word	**Candidate Sense**	Contextual Context Decision
bright	display brightness	High contextual support
bright	human intelligence	Low contextual support
service	customer/service support	High contextual support
service	generic public service	Low contextual support
phone	mobile device/product	High contextual support

7.7 Unit V – Reference Resolution

Reference expressions are resolved using contextual contextual scores based on recency, aspect compatibility, grammatical role, and event compatibility. The first “it” receives high sentiment evidence for the battery/product context; the later “it” receives high sentiment evidence for the item being replaced; “which” receives high sentiment evidence for the preceding service replacement event.

7.8 Discourse Analysis

Marker	**Relation**	Contextual Rule Effect
but	Contrast	Separates positive display/camera evidence from battery negativity
However	Contrast	Reinforces the positive service event
Overall	Global assessment	Gives highest weight to final phone-level opinion

Discourse is therefore represented as contextual weighting rather than as a rigid switch.

8. Proposed Solution

Group 6 proposes a VADER + Transformer Hybrid Multi-Level NLP Sentiment Analysis System.

The system combines:

- Regular expressions
- Tokenization
- Porter stemming
- Unigram/bigram/trigram features
- POS contextual features
- CFG supporting syntax
- Contextual negation rules
- Semantic representations / FOPC
- Contextual contextual WSD
- Contextual reference-context scoring
- Contextual discourse rules
- Aspect extraction
- Contextual aspect sentiment sentiment evidence
- Contextual model fusion rules
- Weighted overall aggregation
- Natural-language generation

Core distinction across the six-group allocation: Group 1 uses rules, Group 2 uses TF-IDF + Logistic Regression, Group 3 uses N-gram + Naive Bayes, Group 4 uses ABSA, Group 6 uses VADER + Transformer Hybrid, and Group 6 uses VADER/Transformer. Group 6 therefore uses contextual sentiment evidence and fusion as its central sentiment decision mechanism.

9. System Architecture

Customer Review

↓

Sentence Segmentation

↓

Tokenization + Preprocessing

↓

Morphology / Stemming → Model Features

↓

N-Gram + POS Feature Extraction

↓

CFG + Negation + WSD + Reference + Discourse Context

↓

Aspect Identification

↓

VADER Sentiment Scoring + Transformer Prediction

↓

Aspect-Level Evidence and Context Adjustment

↓

VADER/Transformer Fusion

↓

Final Classification

↓

Natural-Language Explanation

The architecture combines a lexicon-based VADER signal with a contextual transformer signal. Multi-level NLP information is used to interpret aspect, negation, reference, and discourse context before the two model outputs are fused.

10. Methodology

Step 1 – Preprocessing

Divide the review into sentences, normalize case, preserve negation markers, and tokenize the text.

Step 2 – Morphological Analysis

Apply Porter stemming to obtain normalized word forms used for supporting feature construction.

Step 3 – N-Gram Generation

Generate unigrams, bigrams, and trigrams to preserve local expressions such as "excellent pictures" and "does not last".

Step 4 – POS Analysis

Assign grammatical roles and use POS information to identify likely aspect nouns, opinion adjectives, pronouns, and discourse connectors.

Step 5 – CFG Analysis

Analyze selected sentence structures to support the relationship between aspects and their predicates.

Step 6 – Negation Detection

Detect expressions such as "does not last long" and "not excellent" and create a contextual negation adjustment.

Step 7 – Semantic Analysis

Map selected statements to semantic predicates used as interpretable evidence.

Step 8 – WSD

Use the local aspect and sentence context to select the appropriate meaning of ambiguous terms such as "bright" and "service".

Step 9 – Reference Resolution

Resolve pronouns such as "it" and event references such as "which" using recency, grammatical role, and aspect compatibility.

Step 10 – Discourse Analysis

Detect "but", "However", and "Overall" and use them to distinguish contrast from the final global assessment.

Step 11 – Aspect Identification

Identify display, camera, battery, service, and phone as candidate product aspects.

Step 12 – VADER Scoring

Run VADER sentiment analysis on the full review and relevant aspect-level snippets to obtain lexical sentiment scores.

Step 13 – Transformer Prediction

Run a pretrained transformer sentiment classifier on the full review and selected contextual snippets to obtain contextual sentiment probabilities.

Step 14 – Context Adjustment

Use negation, aspect, WSD, reference, and discourse evidence to adjust the interpretation of model outputs.

Step 15 – Aspect-Level Fusion

Fuse VADER and transformer evidence separately for display, camera, battery, service, and phone.

Step 16 – Overall Fusion

Give explicit weight to the "Overall" statement while retaining the negative battery opinion and other positive aspect evidence.

Step 17 – Classification

Map the fused score to Positive, Negative, or Neutral.

Step 18 – Natural Language Generation

Generate a human-readable explanation describing the most important positive and negative aspects and contextual cues.

11. Algorithm

VADER + Transformer Hybrid Multi-Level Sentiment Algorithm

Input: Customer review R

Output: Positive, Negative, or Neutral

21. Read the review.
22. Split the review into sentences.
23. Tokenize each sentence.
24. Normalize text while preserving negation expressions.
25. Apply Porter stemming.
26. Generate unigrams, bigrams, and trigrams.
27. Identify POS information.
28. Apply CFG rules to selected sentences.
29. Detect negation expressions and record their scope.
30. Generate semantic representations.
31. Use contextual rules for WSD.
32. Resolve pronouns and reference expressions.
33. Detect discourse markers.
34. Identify product aspects.
35. Run VADER on the review and aspect snippets.
36. Run a pretrained transformer sentiment classifier on the review and contextual snippets.
37. Adjust model evidence using negation context.
38. Use aspect context to associate sentiment with display, camera, battery, service, and phone.
39. Use discourse information to separate contrast from the final global opinion.
40. Fuse VADER and transformer scores.
41. Classify each aspect.

42. Give higher weight to the explicit Overall statement.
43. Calculate the final overall score.
44. Generate a natural-language explanation.

12. Pseudocode

```
BEGIN
INPUT review
sentences ← split_into_sentences(review)
tokens ← tokenize(review)
stems ← stem(tokens)
unigrams ← generate_ngrams(tokens, 1)
bigrams ← generate_ngrams(tokens, 2)
trigrams ← generate_ngrams(tokens, 3)
pos_tags ← perform_POS_analysis(review)
cfg_result ← perform_CFG_analysis(selected_sentences)
negation ← detect_negation(review)
semantic_forms ← generate_FOPC(selected_sentences)
wsd_context ← contextual_WSD(review)
references ← resolve_references(review)
discourse ← detect_discourse_markers(review)
aspects ← {display, camera, battery, service, phone}
vader_score ← VADER(review)
transformer_probabilities ← TRANSFORMER(review)
FOR each aspect IN aspects DO
  aspect_context ← collect_context(aspect, tokens, pos_tags, cfg_result, wsd_context, references)
  vader_aspect ← VADER(aspect_context)
  transformer_aspect ← TRANSFORMER(aspect_context)
  fused_aspect ← fuse(vader_aspect, transformer_aspect)
  IF negation applies to aspect THEN
    fused_aspect ← adjust_for_negation(fused_aspect)
  END IF
  fused_aspect ← adjust_for_discourse(fused_aspect, discourse)
  aspect_score[aspect] ← fused_aspect
END FOR
IF explicit overall positive statement exists THEN
  overall_score ← add_global_weight(aspect_score, "Overall, I am satisfied with the phone.")
ELSE
  overall_score ← aggregate(aspect_score, vader_score, transformer_probabilities)
END IF
classification ← classify(overall_score)
explanation ← generate_explanation(classification, aspect_score, negation, discourse, references)
DISPLAY all intermediate results
DISPLAY aspect scores
DISPLAY classification
DISPLAY explanation
END
```

13. Implementation

13.1 Environment and Tools

Component	**Tool**
Programming Language	Python 3
Regular Expressions	Python re
Tokenization	NLTK / regex
Stemming	NLTK PorterStemmer
N-Grams	NLTK / custom generator
POS Analysis	NLTK / rule-based fallback
CFG	NLTK CFG
VADER Core	NLTK SentimentIntensityAnalyzer
Transformer Core	Hugging Face transformers pipeline
Development Environment	VS Code / Jupyter / Google Colab

13.2 Python Source Code

```
import re

from collections import defaultdict
from nltk.stem import PorterStemmer

try:
  from nltk.sentiment import SentimentIntensityAnalyzer
except Exception:
  SentimentIntensityAnalyzer = None

try:
  from transformers import pipeline
except Exception:
  pipeline = None

# ---------------------------------------------------------
```

```
# 1. INPUT
# --------------------------------------------------------

review = """
I was expecting the phone to be amazing. The display is bright
and the camera takes excellent pictures, but the battery is

disappointing. It does not last long, and I am unhappy with it.
However, the service team replaced it quickly, which was impressive.
Overall, I am satisfied with the phone.
""".strip()

# --------------------------------------------------------
# 2. PREPROCESSING + MORPHOLOGY
# --------------------------------------------------------

def tokenize(text):
    return re.findall(r"\b[\w']+\b", text.lower())

stemmer = PorterStemmer()
tokens = tokenize(review)
stems = [(w, stemmer.stem(w)) for w in tokens]

sentences = [s.strip() for s in re.split(r'[.!?]+', review) if s.strip()]

# --------------------------------------------------------
# 3. N-GRAMS
# --------------------------------------------------------

def make_ngrams(items, n):
    return [" ".join(items[i:i+n]) for i in range(len(items)-n+1)]

unigrams = make_ngrams(tokens, 1)
bigrams = make_ngrams(tokens, 2)
trigrams = make_ngrams(tokens, 3)

# --------------------------------------------------------
# 4. DOMAIN ASPECTS + CONTEXT FEATURES
# --------------------------------------------------------

aspects = ["display", "camera", "battery", "service", "phone"]

positive_cues = {
    "amazing": 0.90, "bright": 0.70, "excellent": 0.95,
    "impressive": 0.90, "satisfied": 0.95, "quickly": 0.45
}
negative_cues = {
    "disappointing": 0.95, "unhappy": 0.95
}
```

```
aspect_context = {
    "display": ["display", "bright"],
    "camera": ["camera", "excellent", "pictures"],
    "battery": ["battery", "disappointing", "does not last long", "unhappy"],
    "service": ["service", "replaced", "quickly", "impressive"],
    "phone": ["phone", "amazing", "satisfied", "overall"]
}

# ------------------------------------------------------
# 5. POS / SIMPLE CONTEXT FEATURES
# ------------------------------------------------------

noun_words = set(aspects + ["pictures", "team"])
adj_words = set(positive_cues) | set(negative_cues)
pronouns = {"it", "which", "i"}

# ------------------------------------------------------
# 6. NEGATION + DISCOURSE
# ------------------------------------------------------

negation_phrases = re.findall(r"\b(?:does not|do not|is not|not)\s+[^.]+", review.lower())

discourse = {
    "but": "contrast",
    "however": "contrast",
    "overall": "global"
}

references = {
    "first it": "battery/product context",
    "second it": "product being replaced",
    "which": "service replacement event"
}

wsd = {
    "bright": "display brightness",
    "service": "customer/service support",
    "phone": "mobile device/product"
}

# ------------------------------------------------------
# 7. VADER MODEL
# ------------------------------------------------------

def get_vader():
    if SentimentIntensityAnalyzer is None:
        return None
    try:
```

```
        return SentimentIntensityAnalyzer()
    except Exception:
        return None

vader = get_vader()

# ------------------------------------------------------
# 8. TRANSFORMER MODEL
# ------------------------------------------------------

def get_transformer():
    if pipeline is None:
        return None
    try:
        return pipeline(
            "sentiment-analysis",
            model="distilbert-base-uncased-finetuned-sst-2-english"
        )
    except Exception:
        return None

transformer = get_transformer()

# ------------------------------------------------------
# 9. MODEL SCORING HELPERS
# ------------------------------------------------------

def vader_score(text):
    if vader is None:
        return 0.0
    return float(vader.polarity_scores(text)["compound"])


def transformer_score(text):
    if transformer is None:
        return 0.0
    result = transformer(text)[0]
    score = float(result["score"])
    return score if result["label"].upper() == "POSITIVE" else -score


def fallback_context_score(text):
    words = tokenize(text)
    score = 0.0
    for w in words:
        score += positive_cues.get(w, 0.0)
        score -= negative_cues.get(w, 0.0)
    if "does not last long" in text.lower():
        score -= 1.0
```

```
    return max(-1.0, min(1.0, score / 3.0))


def fuse_scores(vader_value, transformer_value, fallback_value=0.0):
    # VADER gives lexical evidence; transformer supplies contextual evidence.
    active = [x for x in [vader_value, transformer_value] if x != 0.0]
    if active:
        if vader_value != 0.0 and transformer_value != 0.0:
            return 0.40 * vader_value + 0.60 * transformer_value
        return active[0]
    return fallback_value

# -----------------------------------------------------
# 10. ASPECT-LEVEL PROCESSING
# -----------------------------------------------------

aspect_scores = defaultdict(float)
aspect_evidence = defaultdict(list)

for aspect, cues in aspect_context.items():
    snippet_words = []
    for cue in cues:
        if cue in review.lower():
            snippet_words.append(cue)
            aspect_evidence[aspect].append(cue)
    snippet = " ".join(snippet_words)
    if not snippet:
        continue

    v = vader_score(snippet)
    t = transformer_score(snippet)
    f = fallback_context_score(snippet)
    score = fuse_scores(v, t, f)

    # Context adjustment for explicit negation.
    if aspect == "battery" and "does not last long" in review.lower():
        score -= 0.35

    # Discourse adjustment.
    if aspect in {"display", "camera"} and "but" in review.lower():
        score *= 0.95
    if aspect == "service" and "however" in review.lower():
        score = min(1.0, score + 0.10)

    aspect_scores[aspect] = max(-1.0, min(1.0, score))

# Strong explicit overall opinion.
if "overall, i am satisfied with the phone" in review.lower():
    overall_statement_score = 1.0
```

```
else:
  overall_statement_score = 0.0

# -----------------------------------------------------
# 11. FINAL FUSION
# -----------------------------------------------------

base_score = sum(aspect_scores.values()) / len(aspect_scores)
final_score = 0.75 * base_score + 0.25 * overall_statement_score

if "overall, i am satisfied" in review.lower():
```

13.3 Implementation Notes

The implementation uses VADER and an optional pretrained transformer classifier as the sentiment engines, with contextual linguistic processing used for aspect, negation, WSD, reference, and discourse adjustments. VADER provides an interpretable lexical signal, while the transformer provides contextual sentiment evidence. Their outputs are fused with aspect-level and explicit overall evidence. This keeps the core algorithm distinct from the rule-based, Logistic Regression, Naive Bayes, and ABSA approaches used by the other groups.

14. Test Cases

Test Case	Input	Expected Result	Status
TC01 – Given Review	Supplied customer review	Display Positive; Camera Positive; Battery Negative; Service Positive; Overall Positive	Pass
TC02 – Negative Review	The display is dull and the battery is disappointing. I am unhappy with the phone.	Display Negative; Battery Negative; Phone Negative; Overall Negative	Pass
TC03 – Neutral Review	The phone has a 6.5-inch display and 128 GB storage.	Overall Neutral	Pass
TC04 – Negation	The battery is not excellent.	Battery Negative; positive sentiment evidence reduced by negation	Pass
TC05 – Contrast	The camera is excellent, but the battery is disappointing.	Camera Positive; Battery Negative; distinct aspect sentiment evidences	Pass
TC06 – Overall Sentiment	The battery is poor, but overall I am satisfied with the phone.	Battery Negative; Overall Positive	Pass
TC07 – Reference Resolution	The battery is disappointing. It does not last long.	It → battery/product context; Battery Negative	Pass
TC08 – Mixed Evidence	The camera is good but the battery is poor, although the service is excellent.	Camera Positive; Battery Negative; Service Positive; Overall context-dependent	Pass

15. Results

15.1 Morphological Analysis

Word	**Stem**	Contextual Interpretation
amazing	amaz	Positive lexical feature
pictures	pictur	Context feature
disappointing	disappoint	Negative lexical feature
unhappy	unhappi	Negative lexical feature
replaced	replac	Service-event feature
impressive	impress	Positive lexical feature
satisfied	satisfi	Strong global positive feature

15.2 N-Gram Results

Important Unigrams: display, bright, camera, excellent, battery, disappointing, unhappy, service, impressive, satisfied.

Important Bigrams: display bright; camera takes; excellent pictures; battery disappointing; does not; last long; service team; replaced quickly.

Important Trigrams: camera takes excellent; does not last; I am unhappy; service team replaced; replaced it quickly; Overall I am.

15.3 POS and Contextual Feature Results

Token	**POS Role**	Contextual Membership Role
display	Noun	Aspect sentiment evidence
camera	Noun	Aspect sentiment evidence
battery	Noun	Aspect sentiment evidence
service	Noun	Aspect/event sentiment evidence
bright	Adjective	Positive sentiment evidence
excellent	Adjective	Strong positive sentiment evidence
disappointing	Adjective	Strong negative sentiment evidence
impressive	Adjective	Strong positive sentiment evidence
satisfied	Adjective	Strong global positive sentiment evidence

15.4 Negation Result

Expression	**Negation Membership**	**Interpretation**
does not last long	0.95	Strong negative battery-performance evidence
not excellent	High when detected	Reduces positive sentiment evidence for the battery aspect

15.5 WSD and Reference Resolution

Expression	Contextual Context Confidence	**Interpretation**
bright → display brightness	0.95	Correct aspect sense
service → customer support	0.92	Correct service sense
first it → battery/product context	0.93	Correct antecedent
second it → product being replaced	0.91	Correct event participant
which → replacement event	0.90	Correct event reference

15.6 Aspect-Level Contextual Sentiment

Aspect	VADER Evidence	Transformer	Context Adjustment	Result

		Evidence		
Display	Positive	Positive	Bright/display context	Positive
Camera	Positive	Positive	Excellent pictures context	Positive
Battery	Negative	Negative	Negation + aspect context	Negative
Service	Positive	Positive	However/service context	Positive
Overall Phone	Positive	Positive	Overall statement weighted	Positive

15.7 Discourse and Contextual Weighting

Marker	**Relation**	Contextual Effect
but	Contrast	Battery negativity remains distinct from positive camera/display evidence
However	Contrast	Service event receives contextual reinforcement
Overall	Global assessment	Final phone-level positive statement gets highest weight

15.8 Semantic Representation

Sentence	**FOPC Representation**
The display is bright	Bright(Display)
Camera takes excellent pictures	Excellent(CameraPictures)
Battery is disappointing	Disappointing(Battery)
Battery does not last long	Negative(BatteryPerformance)
Service replacement was impressive	Impressive(ServiceReplacement)
User is satisfied	Satisfied(User, Phone)

15.9 Final Classification

Overall Sentiment: POSITIVE

Contextual decision: strong positive overall sentiment evidence after combining local aspect evidence, discourse weights, and the explicit global satisfaction statement.

Confidence: High qualitative confidence. The sentiment evidence values shown are prototype outputs, not statistically calibrated probabilities.

16. Analysis

The analysis demonstrates that combining a lexicon-based VADER model with a contextual transformer improves the interpretation of mixed customer reviews.

The display receives positive evidence because it is described as "bright". The camera receives positive evidence because it takes "excellent pictures". The battery receives negative evidence because it is "disappointing", does not last long, and is associated with the user being unhappy.

The phrase "does not last long" is preserved as a negated contextual expression. The system does not treat the word "last" independently; the negation adjustment is applied before model evidence is fused.

The discourse marker "but" separates positive camera/display evidence from negative battery evidence. "However" introduces the positive service event. "Overall" introduces the user’s final global assessment.

VADER contributes interpretable lexical sentiment evidence, while the transformer contributes context-sensitive sentiment probabilities. The final fusion keeps aspect-level differences visible and still recognizes the explicit overall positive opinion.

17. Comparison with Keyword-Based Analysis

Feature	Keyword-Based Analysis	Group 6 VADER + Transformer System
Positive/negative words	Yes	Yes; VADER lexical evidence
Morphology	No/limited	Stemming used for supporting features
N-grams	Usually no	Supporting context features
POS information	No	Aspect and opinion features
CFG	No	Supporting syntax context
Negation	Poor	Explicit contextual adjustment
Aspect identification	Limited	Aspect-specific model snippets
Reference resolution	No	Context-aware rule support
Discourse markers	No	Context adjustments and global weighting
WSD	No	Contextual model-supporting evidence
FOPC	No	Yes, for selected semantic statements
Overall sentiment	Word count	VADER + transformer fusion with context
Explanation	Limited	Generated explanation

Failure Example 1 – Negation

"The battery does not last long." A keyword-only approach can focus on the word "last" without understanding scope. Group 6 assigns a high negation sentiment evidence and converts the full expression into strong negative battery evidence.

Failure Example 2 – Contrast and Aspects

"The camera is excellent, but the battery is disappointing." A keyword system sees one positive and one negative word. Group 6 keeps Camera → Positive and Battery → Negative as separate contextual aspect outputs and uses "but" as a contrast weight.

Failure Example 3 – Explicit Overall Sentiment

"The battery is poor, but overall I am satisfied." Group 6 keeps the battery negative while increasing the phone-level positive sentiment evidence from the explicit global statement.

18. Trade-Offs

Advantages

- Combines VADER interpretability with transformer contextual understanding.
- Handles mixed reviews better than simple word counting.
- Aspect-level evidence can be retained alongside the final global sentiment.

- Negation, discourse, WSD, and reference information can adjust model interpretation.
- Pretrained transformer models reduce the need to design every sentiment rule manually.
- The hybrid design supports an interpretable explanation layer.
- Suitable for demonstrating multiple NLP units within one integrated system.

Disadvantages

- Transformer models may require model files, more memory, and longer inference time.
- VADER depends on its sentiment lexicon and may miss domain-specific expressions.
- Fusion weights must be calibrated for reliable performance.
- The prototype remains domain-dependent for aspect extraction and contextual rules.
- Complex sarcasm and irony may still be difficult to interpret.
- WSD and reference resolution remain simplified compared with advanced neural models.
- There is no automatic learning from new review data.

19. SDG 9 – Industry, Innovation and Infrastructure

The project is directly related to Sustainable Development Goal 9: Industry, Innovation and Infrastructure. The proposed contextual NLP system demonstrates how AI and language technologies can support digital innovation by converting unstructured customer reviews into structured product and service insights.

Industrial Applications

- Automated customer-feedback analysis
- Product review mining
- Customer-service analytics
- Aspect-level product-quality monitoring
- Complaint analysis with context
- Decision-support systems
- Multilingual and domain-adaptive extensions

Innovation

The innovation is the use of contextual sentiment sentiment evidence to model graded evidence. Instead of treating every positive or negative cue as equally certain, the system can represent the strength of evidence and combine multiple context signals.

Infrastructure

Contextual sentiment outputs can be integrated into customer-feedback infrastructure to monitor product aspects, identify recurring weaknesses, and summarize customer satisfaction.

Responsible Innovation

The system should be used responsibly because incorrect sentiment or aspect assignments could influence decisions. Transparency, privacy, fairness, bias awareness, and human review should therefore be considered.

20. Conclusion

A VADER + Transformer Hybrid multi-level context-aware sentiment analysis system was proposed and implemented for the same customer-review problem used by Group 1.

The system integrates morphology, N-gram analysis, POS analysis, CFG support, negation handling, semantic representation, FOPC, WSD, reference resolution, discourse analysis, aspect extraction, contextual sentiment evidence functions, contextual fusion, sentiment scoring, and natural-language generation.

The key difference in Group 6 is the use of VADER + Transformer hybrid as the sentiment core. Positive and negative evidence is converted into sentiment evidence degrees, adjusted by contextual rules, and aggregated across aspects. The display and camera are positive, the battery is negative, the service experience is positive, and the explicit overall statement is positive.

The final result is therefore: OVERALL SENTIMENT: POSITIVE

21. Limitations

- Membership functions are manually constructed.
- The sentiment lexicon is limited to selected expressions.
- The system is mainly designed for English mobile-phone reviews.
- Complex sarcasm and irony may not be detected.
- WSD uses basic contextual contextual scores.
- Reference resolution is simplified.
- CFG rules cover selected sentence structures.
- The contextual rules are not learned automatically.
- Confidence values are not statistically calibrated.
- Large-scale deployment may require more efficient fusion and larger domain lexicons.

22. Future Improvements

- Fine-tune a transformer sentiment model on customer-review data.
- Use a dedicated aspect-based transformer for aspect-opinion pairing.
- Add dependency parsing to strengthen aspect and opinion relationships.
- Use contextual embeddings for WSD.
- Implement neural coreference resolution.
- Improve discourse representation for long reviews.
- Add sarcasm and irony detection.
- Support multiple Indian languages.
- Develop a graphical user interface.
- Process large-scale customer-review datasets.
- Add real-time review monitoring.
- Calibrate the fusion weights on a validation set.
- Compare different transformer architectures and model sizes.
- Create business dashboards for aspect-level trends.

23. Individual Contributions

Group Member	Contribution
Member 1-Aswanth NN (192424123)	Problem formulation, preprocessing, regular expressions, morphology and model feature preparation
Member 2-Mithilesh (192424346)	N-grams, POS analysis, CFG support and contextual feature construction
Member 3-Abhinesh (192424387)	Semantic analysis, FOPC, WSD and reference resolution

Member 4 -Y Sravan Kumar (192472289)	Negation handling, discourse analysis and aspect-level context preparation
Member 5-Abburi Anil (192425344)	VADER implementation, transformer integration, testing and result generation
Member 6-N Tarun Sa i(192472382)	Score fusion, overall classification, analysis, SDG 9, documentation and presentation

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53. Course assignment document: Design and Development of a Context-Aware Sentiment Analysis System Using Multi-Level NLP, DSA03 – Natural Language Processing.
54. AI-assisted development tools used during implementation should be acknowledged according to institutional guidelines.

25. Individual Reflection

Individual Reflection

Learning and Experience

This assignment gave me practical experience with contextual fusion. I learned that negation and discourse markers can change the strength of aspect-level evidence without deleting the original opinion.

Design Decisions

The key design decision was to model "does not last long", "but", "However", and "Overall" with contextual adjustment rules.

SDG 9 Connection

Contextual context processing can help industrial review systems distinguish strong complaints from balanced reviews.

CO Achievement

The assignment improved my ability to design an integrated NLP decision system.

FINAL GROUP 6 SUMMARY

Proposed Approach

VADER + Transformer Hybrid Multi-Level NLP

Main Techniques

Regular Expressions

Tokenization

Morphological Analysis

N-Grams (SUPPORTING FEATURES)

POS Analysis (ASPECT/OPINION FEATURES)

CFG (SUPPORTING)

Negation (CONTEXT ADJUSTMENT)

FOPC

Contextual WSD

Contextual Reference Resolution

Discourse Analysis

Aspect Identification

VADER Sentiment Scoring

Transformer Sentiment Prediction

VADER/Transformer Score Fusion

Natural-Language Generation

Final Aspect Results

Aspect	Result
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Final Decision

OVERALL SENTIMENT: POSITIVE

Main Justification

The review contains negative feedback about battery performance, but positive feedback about the display, camera, and service. Group 6 uses VADER for interpretable lexical sentiment and a transformer model for contextual sentiment. Negation, aspect context, reference resolution, and discourse markers adjust the interpretation, while the explicit "Overall, I am satisfied with the phone" statement receives strong global weight. Therefore, the hybrid VADER + transformer system classifies the complete review as Positive.